Is this reading the one the wall was expected to show?

A grey-box NeuralProphet decomposition, rolling expectation and three-scale monitor of the station 02 inclination, 2018 to 2026

Study report · `studies/05_greybox_monitoring/`

Contents

1 Introduction

A reading arrives from the wall every twenty minutes: the inclination of the instrument at station 02, the air temperature and relative humidity beside it, and, since 21 February 2025, the temperature of the stone and the solar radiation falling on it. The question a person watching that stream must be able to answer is operational rather than diagnostic. Given the environment measured at this instant, is this inclination the one the wall was expected to show, and if it is not, which kind of departure is it? Study 04 posed that question on the window after the long outage of 2022 to 2023 and answered part of it. It also recorded its own weak points: an annual term that was not identifiable on 3.2 interrupted cycles, and a trend that changed sign across the thirds of its window; a conformal interval calibrated once, eleven months before the record it was then asked to cover, that reached 67.7 % coverage against a nominal 90 %; a residual whose lag-one autocorrelation of 0.995 pushed the control limit to 14.75 standard deviations and left the detector blind to every slow mechanism; and a phase-change injection scaled by the leftover daily seasonal term rather than by the wall's own daily response, which overstated that blind spot by roughly five times. This study keeps what Study 04 established and changes what it got wrong.

Four studies precede it, and their conclusions are inherited rather than re-litigated. Study 01 read the raw archive, decided which recorded values are measurements, and exported the inclination product used here and no other: compensated for temperature with the manufacturer's linear correction, anchored once across the whole record, cleaned of impulsive noise, and accompanied by a flag that marks every value the cleaning replaced. It also established that the two instruments installed at station 02 are continuous across the changeover of 21 February 2025, and that the excursions of summer 2026 are an instrument or power artefact rather than a structural or environmental event. Study 02 characterised the two external sources, the town's ground station and the ERA5 reanalysis, against the on-structure channels: the station is the closer proxy, ERA5 the more transferable one, and both report global horizontal radiation, the quantity the wall's own pyranometer measures. Study 03 measured the couplings: air temperature drives the inclination instantaneously, with a diurnal-band gain near $-2.8\,\mathrm{mdeg/°C}$ on the on-structure channel; radiation acts with a delay of about one hour; humidity follows the temperature cycle inversely rather than acting as a mechanism of its own; and the wall temperature is an internal state that leads the deformation and must not be used predictively at its measured lead. Study 04 settled the modelling choices this study keeps: no autoregression in the decomposition, because it absorbs the diurnal structure the regressors should explain; changepoints placed on covered time; the level decomposed and monitored rather than scored as a forecast; contemporaneous regressors in the expectation treated as legitimate inputs rather than as leakage; and no value of the target ever imputed, with the library's own gap filling switched off.

Two boundaries hold throughout. The thermal compensation is taken as given: it is the manufacturer's procedure, it has been validated against the user's numerical model of the wall, and every gain this study learns is therefore a post-compensation wall response and is labelled as such. No fit is made on the raw channel. Forecasting is out of scope, because Study 04 answered it: the skill of a forecast comes from the response's own memory, and the environment adds nothing distinguishable from zero beyond six hours.

Within those boundaries the study asks five questions. What is the record made of, over eight years: trend, annual cycle, daily cycle and the response to each measured driver, with the share of variance each carries and the gain each learns, on three regressor sets? Does the wall answer its drivers with a delay or an inertia that the twenty-minute grid can resolve, when the response is learned rather than imposed? Is this reading the expected one, given a rolling expectation and an interval whose coverage is measured as a function of how stale the last refit is? Which departure does each of three control charts catch, at a false-alarm budget stated in advance, when the injected departures are sized by the wall's measured daily response? And what happened across each outage, when the level expected at resumption from the proxies that kept recording is set against the level observed?

2 The record and the three regressor sets

The study window is the whole station 02 record, from 26 July 2018 to the end of the archive in August 2026, on the instrument's native grid of one reading every twenty minutes: 212,251 slots. Unlike Study 04, which worked on the stretch after the long outage of 2022 to 2023, this study uses every usable annual cycle, because an annual term cannot be identified on three broken years, and a trend estimated on such a window disagreed with itself in sign (Section 3, D1).

The response is joined by three regressor sets, each mapping the same three roles, air temperature, relative humidity and solar radiation, to a different source (D2). The on-structure set reads air temperature and humidity from the wall's own package, joined across the two instrument eras; the station set reads the town's ground station; the ERA5 set reads the reanalysis. The on-structure set borrows the station's radiation, because the wall's pyranometer did not exist before February 2025 and Study 01 condemned 161 of its days after that; every table that reports the on-structure set carries that borrowing. The two external sources are hourly or half-hourly and are brought onto the twenty-minute grid by linear interpolation between consecutive native observations, never across a longer gap, and the ERA5 radiation, an accumulation over the preceding hour, is centred on the half-hour before it is interpolated (D13). Regressor dropouts of at most two hours are then filled by linear interpolation and flagged; the target is never filled.

Table 1 gives what each set actually holds over the window. The inclination returns an accepted value in 67.7 % of the slots. The on-structure air temperature and humidity reach 71.1 %, with about two per cent of their accepted slots filled across short dropouts; the station's channels reach 91.7 to 94.7 %, with under one per cent filled; ERA5 covers 99.2 % of the window with nothing to fill. The radiation of the on-structure and station sets is the same column and has the same coverage.

Table 1: Accepted and filled slots and coverage per regressor set and role over the study window, on the twenty-minute grid, with the target as the last row. The on-structure set's radiation is the station's column, borrowed. Read from `GM_01_window_coverage.csv`.

Set	Role	Accepted slots	Filled slots	Coverage
`str`	`tair`	150,844	3,049	71.1%
`str`	`rh`	150,844	3,045	71.1%
`str`	`sr`	201,037	1,090	94.7%
`gs`	`tair`	200,862	1,145	94.6%
`gs`	`rh`	194,675	1,140	91.7%
`gs`	`sr`	201,037	1,090	94.7%
`era5`	`tair`	210,473	0	99.2%
`era5`	`rh`	210,473	0	99.2%
`era5`	`sr`	210,455	0	99.2%
`target`	`inc_comp_cleaned`	143,795	0	67.7%

Table 2 shows how the target's missing time is distributed. Of the 3,302 gaps in the accepted inclination, 2,616 last an hour or less and 595 between one and six hours; together they hold 2,679 of the 22,819 missing hours. Nine gaps longer than seven days hold 19,234 hours, 84 % of the missing time. These are gaps in the accepted values rather than in the raw archive, and they are therefore longer than the outages the data-quality report lists: two interruptions of three weeks and one of eight in the autumn of 2018 and the first half of 2019, when the legacy instrument's readings were condemned; 57 days in the summer of 2020, around an archive outage of eleven; a single gap of 437.6 days from 9 April 2022 to 21 June 2023, in which the two outages of 2022 and the condemned weeks before and between them merge; and the outages of 2024, 2025 and 2026, of 41.5, 42.7, 17.8 and 104 days. Nothing is written into any of them. They are what the rolling expectation must be refitted around, and the ones that fall in the monitored period are the hypotheses that Section 11 tests.

Table 2: The target's missing time by gap duration: how many gaps of each class the accepted inclination has, and how many hours they hold. Read from `GM_02_gap_inventory.csv`.

Gap class	Gaps	Hours
`<=1h`	2,616	1,180
`1-6h`	595	1,499
`6-24h`	81	881
`1-7d`	1	25
`>7d`	9	19,234

Before anything was joined, each radiation source's clock was checked with Study 02's two-sided test, on the hourly grid the test was designed for (Table 3): the offset of the daily maximum from computed solar noon, and the lag of the cross-correlation against ERA5. The station and ERA5 pass both tests, with solar offsets of 0.11 and 0.44 hours and no cross-correlation lag. ERA5's larger offset is the signature of its hourly accumulation, which is stamped at the end of the hour it describes and which D13 centres before interpolation. The on-structure pyranometer fails the agreement test over its 215 certified days, its solar offset of −0.22 hours sitting against a cross-correlation lag of one hour. No regressor set consumes that channel in the main line; it enters only the current-era ladder of D4, where its clock is part of what the ladder judges.

Table 3: Study 02's clock check on each radiation source, on the hourly grid: the days with a usable solar maximum, the offset of that maximum from computed solar noon with its interquartile range, the cross-correlation lag against ERA5 with its correlation, and whether the two tests agree. Read from `GM_03_clock_check.csv`.

Source	Channel	Days	Solar offset [h]	IQR [h]	Lag [h]	r	Agree
`str`	`sr_str`	215	-0.22	0.72	1.0	0.932	no
`gs`	`sr_gs`	2,469	0.11	0.54	0.0	0.930	yes
`era5`	`sr_era5`	2,965	0.44	0.35	0.0	1.000	yes

Figure 1 draws the record and the three sets on one clock, one record per panel, with the gaps left as gaps.

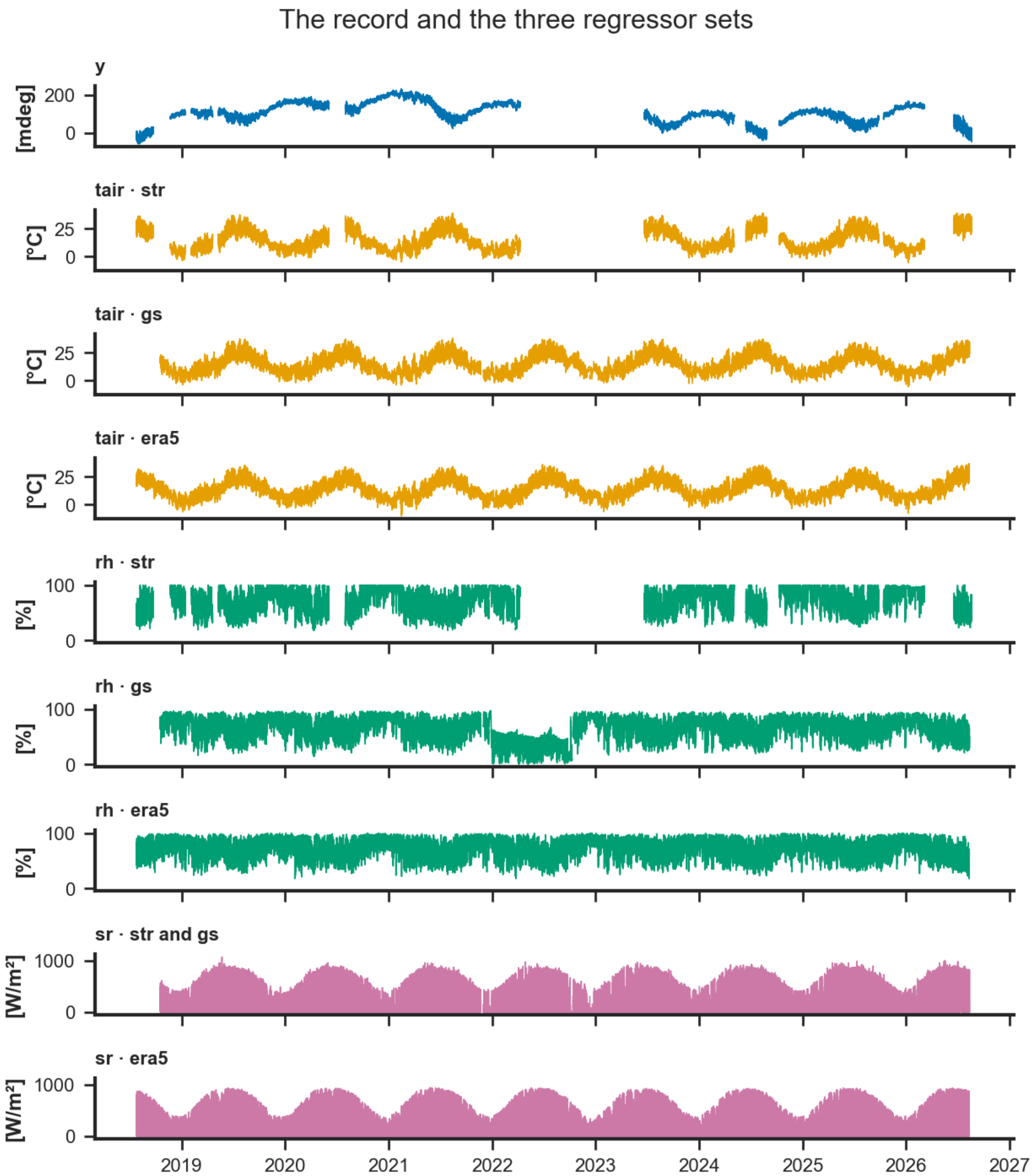

Figure 1: The record and the three regressor sets, 2018 to 2026, on the twenty-minute grid, one record per panel. Top: the compensated, once-anchored, spike-cleaned inclination, its gaps left as gaps. Below: air temperature, relative humidity and global horizontal radiation from each source, the panels of one role sharing their vertical scale. The station's radiation is drawn once and labelled for both the station set and the on-structure set that borrows it; the wall's own pyranometer does not enter the main line.

3 Method

Each subsection states one design decision, the evidence for it, and the alternative it rejects, in the order of the design document. Decisions whose results arrive in later phases, D6 onward, are stated here and reported in their own sections.

3.1 D1 · The whole record, 2018 to the archive end

Study 04's annual term was not identifiable on 3.2 cycles with a 103-day hole, and its trend disagreed with itself in sign across the thirds of its window: +38.6, +4.8 and +45.5 mdeg per year against −2.77 for the window as a whole. The whole record holds five usable annual cycles, 2019, 2020, 2021, 2024 and 2025, compensated and anchored once by Study 01. The instrument change of 21 February 2025 is not modelled: the recorded channel is continuous across it to +1.58 mdeg on thirty-day means, and the compensated gain is stable across both eras. The rejected alternative is a per-era offset, which would plant an arithmetic step the instruments did not record.

3.2 D2 · Three regressor sets through one model specification

One specification is fitted three times, once per regressor set. The on-structure set gives the site result, the station set the best external proxy, and the ERA5 set the transferable arm. The paper narrates two and tables all three. The on-structure set borrows the station's radiation, as Section 2 states. Wall temperature and on-structure radiation stay out of the main line, since they cover 17.6 % and 18.8 % of the window, and are tested on their own window in D4. The rejected alternative, a single set, would not separate what the site can achieve from what a deployment without on-structure instruments could.

3.3 D3 · Study 03's operator is applied outside the model; radiation is global horizontal in every set

Air temperature enters instantaneously. Radiation enters delayed by one hour, the diurnal-band optimum Study 03 measured for all three sources. Both pass through the library's one delay-and-filter operator, and no driver is given a time constant from the level band, where Study 03 showed such constants pin against the grid. Global horizontal irradiance is used in every set, so that the radiation gain is comparable across sets and matches the variable Study 03 screened. ERA5's skin temperature is not loaded: the radiative temperature of a grid cell several kilometres across, averaged over fields, roads and roofs, carries nothing about one stone face that the air temperature and radiation do not already carry. The rejected alternative, letting Model A learn the delay through lagged regressors, is the job of Model B in D9, where the learned response is confronted with the imposed one rather than replacing it.

3.4 D4 · Wall temperature and on-structure radiation, tested as a ladder on the current era

On the window from 21 February 2025 onward, the same specification is refitted rung by rung: the on-structure set as in the main line; on-structure radiation replacing the station's, on the certified days; the wall temperature added at zero delay with a four-hour time constant, the deployable form; and, as a diagnostic row only, the wall temperature at its measured lead of two hours, which is never used for the expectation. Each rung reports held-out error with a paired block-bootstrap increment, variance share, the learned gain beside Study 03's, conformal coverage and width, and residual autocorrelation. The verdict is one sentence per channel on whether a deployment should keep it. The rejected alternative, carrying both channels in the main line, would cost four fifths of the window.

3.5 D5 · The trend stays on in the monitoring fit

Study 04 removed the trend from its monitoring fit so that drift would remain in the residual. The consequence was a residual with lag-one autocorrelation of 0.995, a control limit swept out to 14.75 standard deviations, a 288-day alarm that said only that the wall had moved since calibration, and a detector blind to every slow mechanism. Here the expectation comes from a rolling refit every thirty days with the trend on; over thirty days the trend's extrapolation error is a fraction of a millidegree at the measured rates, and the residual it leaves is stationary enough to chart. Drift is detected by the slow chart of D10 on daily means, and by the sequence of refit trend slopes. Component shares, gains and the parameter figures come from a single attribution fit over the full training window.

3.6 D6 · Harmonic diagnostics before modelling

The Fourier orders of the seasonal terms and the weight curve of the conditional daily term are measured before any model is fitted, on two series: the target, and the residual of a plain least-squares regression of the target on the on-structure drivers, which is what the seasonal terms will actually have to explain. A Lomb–Scargle scan from twelve hours to nine hundred days certifies the periods; the daily cycle's amplitude and phase, fitted day by day, are then fitted against day of year with an annual Fourier series whose order is chosen on held-out years; and the singular values of the daily-by-annual surface say how many weighted daily shapes it takes to reproduce it. Every order and weight used afterwards traces to that table. The results are reported in Section 5.

3.7 D7 · Model A, an additive grey-box without autoregression

Model A is a linear trend with changepoints placed on covered time, an annual term and a daily term as Fourier series of the orders D6 measured, no weekly term, the three regressors of one set as future regressors, quantile regression at 0.05 and 0.95, and no autoregression, with the library's imputation switched off. The daily cycle is fitted as two shapes, a midsummer and a midwinter form, blended by two weights that vary only with the day of year and sum to one, so that the daily cycle's amplitude and phase modulate through the year without a boundary day at which the shape steps. The weight curve is the measured annual modulation of D6, normalised to the unit interval, with a cosine peaking in mid-July as the documented fallback. The conditional term is kept only if it improves the held-out error or carries more than one per cent of the variance, and four boolean seasons are the stated fallback if the smooth weights fail to fit. Training histories shorter than two full annual cycles are not allowed to speak.

3.8 D8 · Uncertainty the library's way, calibrated on a rolling window

Quantile regression inside the model gives a nominal ninety per cent interval; split conformal prediction then calibrates it. Study 04 calibrated once, on a stretch that ended eleven months before the record it covered, and reached 67.7 % coverage. Here each refit calibrates on the most recent six months of its own out-of-sample residuals, so the interval is never calibrated on a quieter era than the one it covers. Coverage, width, Winkler score and pinball loss are reported per set and per days since refit. The stability of the decomposition's components is measured across the library's own chronological folds, each refitted from scratch on the record up to its held-out stretch: a gain or a seasonal amplitude is reported as a finding only if it keeps its sign and its order of magnitude in every fold, and a component that changes sign between folds is not.

3.9 D9 · Model B, a learned impulse response on the on-structure set

The same specification without any autoregression on the target carries air temperature and the station's radiation as lagged regressors with thirty-six lags at twenty minutes, twelve hours of history per driver, at the resolution Study 03's scan could not reach. The learned weight per lag is the impulse response; from it an effective delay and a time constant are read and set beside Study 03's operator for each driver. Agreement validates the operator imposed in Model A; disagreement is reported, and Model A is not re-tuned to hide it. Phase 0 verified that the library accepts lagged regressors without autoregression on the target, so the hourly fallback the design provided for is not needed.

3.10 D10 · Three charts, one per damage mechanism and time scale

Reference statistics come from a fixed window, 2019 to 2021, before the 2022 outages and before the instrument change; the monitored record runs from January 2022. All three charts run on Model A's rolling residual for each set. The fast chart applies an exponentially weighted moving average and a CUSUM to the residual's prewhitened innovations at twenty minutes, for steps, spikes and instrument faults, at a budget of ninety watched days per false alarm. The daily chart applies the same pair to the amplitude and phase of a 24-hour harmonic fitted to each day's residual, for amplitude growth and phase change, at the same budget. The slow chart applies a CUSUM with a small reference value to the daily mean residual, beside the sequence of refit trend slopes, for drift, at a budget of one year. Every fast alarm is cross-checked against the air temperature, humidity and battery channels in the same slot, so that the episode table carries an attribution: environment, instrument, or unattributed.

3.11 D11 · Detectability by mechanism, with injections sized by the wall's daily response

Study 04's three injection shapes, amplitude growth, phase change and drift, are kept with two corrections. The phase injection is sized by the wall's fitted daily response on the injection date rather than by the leftover daily seasonal term, which on the measured numbers means about 5.6 mdeg for a two-hour shift where Study 04 injected 1.07. Injections are placed at several dates across the seasons, so that the detectability field is monotone rather than an artefact of one injection point. Each mechanism is scored on the chart built for it, with the other two reported as well, and an alarm counts only where the uncontaminated record is silent within the response window.

3.12 D12 · Outages as hypotheses, never as data

For each whole-day outage of seven days or more in the monitored period, the model refitted on the data up to the outage predicts the level at resumption from the station and ERA5 sets, which kept recording. The observed mean over the first seven days after resumption, with the restart transient excluded, is compared with the expected level and its conformal interval, giving a level shift with an interval and a verdict per outage and per proxy set. Nothing is written into the gap: an imputed value may serve as history or as hypothesis, never as evidence. Section 2 showed that the accepted target's gaps are longer than the archive's outages; which intervals are bridged is stated in Section 11.

3.13 D13 · The target is never filled; regressor dust is

Rows with a missing target are left out of every fit. Regressor gaps of at most two hours are filled by linear interpolation and flagged in a provenance column, because a regressor is a measured, smooth driver and not the thing being judged; longer regressor gaps leave the row out of the fit for that set only. Proxies are brought to the twenty-minute grid by linear interpolation between consecutive native observations, never across a longer gap, with the ERA5 radiation centred on the half-hour before interpolation. Each source's clock is checked before anything is joined, as Section 2 reports.

3.14 D14 · Every native plot type, redrawn in the project's graphical language

The library's native plots run in the notebook as diagnostics. The report shows the same content redrawn by the project's own figure functions under its binding graphical rules, each redraw reading its data through the library's public methods, the decomposed prediction, the trend and seasonal components, the lagged-regressor weights and the conformal output columns, through one extractor per plot type, and each extractor is tested against the numbers the native plot draws. Phase 0 found that this library version's matplotlib backend returns no figure object from its parameter plot; nothing in the design depends on it.

4 Trend

Model A was fitted once per regressor set for attribution, on the first four fifths of each set's rows, with the last fifth held out as the library's own validation split (D5, D7). On the on-structure set the training window runs from 21 November 2018, the first slot at which all three drivers are present, to 11 March 2025, and holds 109,259 rows; the station and ERA5 sets hold 106,849 and 113,955. Twelve changepoints were placed on covered time, so that none falls inside a gap, and the trend's regularisation was chosen on the held-out tail before the attribution fits were made.

4.1 The regularisation sweep

Table 4 gives the sweep. Without regularisation the held-out mean absolute error is 17.7 mdeg. Every positive candidate gives the same held-out error, 122.347 mdeg on each of the four: the library's penalty on the changepoint deltas, at any of the four strengths tried, drives every delta to zero, and the trend collapses to a single straight line through six and a half years of a record that is not straight. The candidates are in the wrapper's own scale; NeuralProphet multiplies a positive value by 0.001 before it reaches the loss. The study therefore keeps the unregularised trend, and the parameter cell records `TREND_REG = 0.0` with this table as its reason.

Table 4: The trend-regularisation sweep on the on-structure set: the held-out mean absolute error of the last fifth of the rows for each candidate, and the one kept. Read from `GM_05c_trend_reg_sweep.csv`.

Trend regularisation	Held-out MAE [mdeg]	Kept
0.0	17.745	yes
0.5	122.347	no
1.0	122.347	no
2.0	122.347	no
5.0	122.347	no

4.2 The trend on covered time

Figure 2 draws the fitted trend of the on-structure set and the rate of each segment between changepoints; Table 5 lists the rates. The trend carries 60.1 % of the fitted variance on the on-structure set, 60.7 % on the station set and 64.1 % on ERA5 (Table 10), with a peak-to-peak of 116.4, 119.9 and 139.5 mdeg: over six years the level of the compensated inclination moved by more than a hundred millidegrees net of air temperature, humidity, radiation and the annual cycle, and that movement is what the trend holds.

Its shape is not a drift. The level rose through 2019 and 2020, at rates between 17 and 104 mdeg per year over segments of four to six months, to a maximum at the changepoint of 19 November 2020; it then fell through 2021 at 29 to 91 mdeg per year. Across the long gap of 2022–2023 the segment from 5 December 2021 to 26 June 2023 has a rate of −2.2 mdeg per year, which is the straight line between the last data before the gap and the first after it, not a measurement of anything inside it. From mid-2023 the level fell again, at 89 and 49 mdeg per year, to a minimum at the changepoint of 31 March 2024, and rose by 28 mdeg per year through the rest of 2024 before flattening, at 3 mdeg per year, in the first months of 2025.

Two readings of this trend must be kept apart. As a component of the decomposition it is the slow part of the record after the drivers and the annual term, and its rates are the wall's net movement on the scale of months. As evidence about the structure it says nothing on its own: a rate of 100 mdeg per year sustained for five months and then reversed is the signature of a slow, reversible process, and whether that process is structural or a thermal memory of the masonry that the compensation and the instantaneous drivers do not carry is what the monitor of Section 10 and the current-era ladder of Section 9 are built to separate. The mean rate over a fold is not a stable quantity: in the chronological folds of Table 12 it changes sign between folds on the on-structure and station sets, and on ERA5, where it stays positive, it runs from 4.4 to 20.2 mdeg per year, because each fold's training window ends at a different point of the same reversing record. On the first two sets the rule of Section 3 excludes it as a finding; on ERA5 it is reported for what it is, the mean of a reversing record over whatever stretch a fold happens to cover, not a rate of the wall. The segment rates of Table 5 are findings.

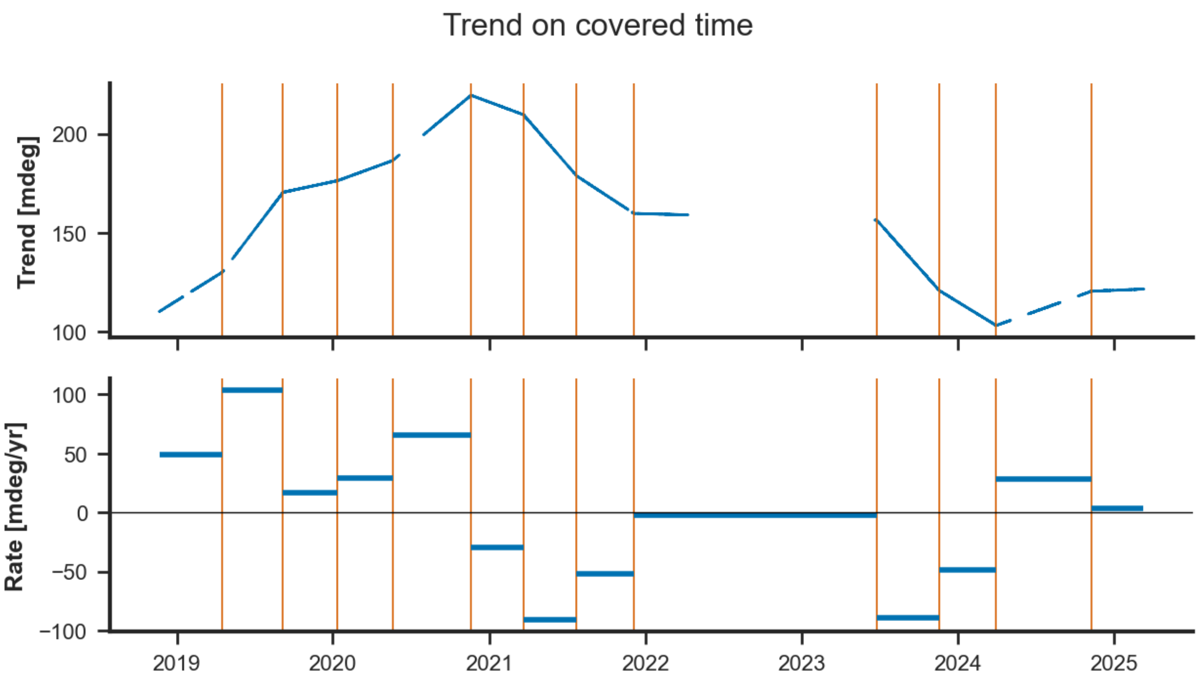

Figure 2: The trend of Model A on the on-structure set over the training window, 2018 to 2025. Top: the fitted trend on covered time, with the twelve changepoints marked. Bottom: the rate of each segment between changepoints, in millidegrees per year. The segment that spans the 2022–2023 gap is the straight line between the data on either side of it.

Table 5: The rate of the fitted trend on each segment between consecutive changepoints of the on-structure fit, from the trend's value at the segment's ends. Read from `GM_04d_trend_rates.csv`.

From	To	Rate [mdeg/yr]
2018-11-21	2019-04-17	49.30
2019-04-17	2019-09-06	103.92
2019-09-06	2020-01-12	16.87
2020-01-12	2020-05-19	28.81
2020-05-19	2020-11-19	65.72
2020-11-19	2021-03-22	-29.29
2021-03-22	2021-07-24	-90.90
2021-07-24	2021-12-05	-51.73
2021-12-05	2023-06-26	-2.18
2023-06-26	2023-11-19	-88.97
2023-11-19	2024-03-31	-48.56
2024-03-31	2024-11-09	28.42
2024-11-09	2025-03-11	3.31

5 Seasonality

5.1 The harmonic diagnostic

The Fourier orders of the seasonal terms and the weight curve of the conditional daily term were fixed before any model was fitted (D6). Two series were examined: the target itself, and the residual of an ordinary least-squares regression of the target on the three drivers of the on-structure set, which is what the seasonal terms will actually have to explain once the regressors have taken their share. That plain fit is a diagnostic device, not a model of the wall; on the levels it attributes the bulk of the daily and annual swing to air temperature; its coefficients are not reported here, and its residual carries whatever the drivers, entering instantaneously, do not.

Each series was scanned with a Lomb–Scargle periodogram, which takes the gaps as they are, on two bands ranked separately: from 0.4 to 3 days for the daily cycle and its harmonics, and from 30 to 900 days for the annual cycle and its harmonics (Table 6). The bands are kept apart because the sub-daily peaks carry a small share of the variance, about 1.1 % for the target's daily cycle and 0.1 % for the residual's, and a single ranking fills every rank with the long-period continuum before it reaches them. On the target the long band certifies the annual cycle at 364.2 days with a normalised power of 0.42, and the semi-annual cycle at 181.7 days with 0.035, inside its 11-day resolution of half a year; the remaining long-band entries at 311, 437 and 600 days are the leakage of the multi-year drift and of the record's gaps rather than cycles. On the residual the annual cycle survives at 352.7 days, within its 44-day resolution of a year, at a power of 0.061, and no semi-annual peak does: the drivers, entering instantaneously, take that component with them. The 900-day entry at the top of the residual's long band is the grid's edge, a third of the record, and is not a period. In the short band both series certify the daily cycle at 1.000 days and its 12-hour companion at 0.500 days; the entries at 0.9996, 1.0005 and 1.0013 days are the daily peak's sidelobes, spaced by the record's resolution and listed because they fall just outside the width within which the scan suppresses a peak's neighbours.

Table 6: Periods certified by the Lomb–Scargle scan on the target and on the regression residual, each band ranked against its own continuum. The resolution is the width around a period that this record cannot tell from the period itself. Read from `GM_04_harmonic_diagnostics.csv`.

Series	Band	Rank	Period [days]	Power
`target`	`short`	1	1.0000	0.0111
`target`	`short`	2	0.9996	0.0023
`target`	`short`	3	1.0005	0.0022
`target`	`short`	4	0.5000	0.0014
`target`	`short`	5	1.0013	0.0009
`target`	`long`	1	364.1820	0.4197
`target`	`long`	2	311.0393	0.0761
`target`	`long`	3	436.8641	0.0568
`target`	`long`	4	600.4277	0.0419
`target`	`long`	5	181.6826	0.0346
`residual`	`short`	1	0.9999	0.0012
`residual`	`short`	2	1.0005	0.0011
`residual`	`short`	3	0.5000	0.0007
`residual`	`short`	4	0.4999	0.0004
`residual`	`short`	5	0.9990	0.0004
`residual`	`long`	1	900.0000	0.0998
`residual`	`long`	2	583.3669	0.0666
`residual`	`long`	3	352.6925	0.0614
`residual`	`long`	4	453.1717	0.0433
`residual`	`long`	5	244.1000	0.0415

For every day with at least 60 of its 72 slots, a 24-hour harmonic and its 12-hour companion were fitted to the diurnal band of each series, giving the daily cycle's amplitude and the hour of its maximum over eight years (Figure 3). Each of those two daily series was then fitted against day of year with an annual Fourier series, the order chosen on held-out years and a higher order accepted only where it lowered the held-out error by at least one per cent (Table 7). On the target the fitted daily amplitude runs from a minimum of 2.8 mdeg in early December (day 342) to a maximum of 17.1 mdeg in late July (day 200), a six-fold swing; on the residual it runs from 2.2 mdeg in mid-April (day 102) to 3.3 mdeg in late July (day 206), a swing of one and a half. The two share the summer peak and not the trough: once air temperature has been removed, the residual's daily cycle is smallest in spring rather than in winter. Both fits take order two, and the residual's fit, the one the conditional daily term consumes, has coefficients $(2.620, -0.292, -0.135, 0.265, 0.268)$ for the constant, the first cosine and sine and the second cosine and sine over a period of 365.25 days; normalised to the unit interval it peaks on day 206. That the residual's daily cycle modulates weakly once air temperature has been removed is itself a finding: most of the annual modulation of the daily swing rides on the driver, and the conditional term of D7 has less to carry than the target's picture suggests. Whether the two weighted shapes still earn their place is decided by the held-out comparison reported in the second half of this section.

Table 7: Annual modulation of the daily cycle's amplitude and hour of maximum, per series: the leave-one-year-out error of each candidate Fourier order and the order chosen. Read from `GM_04_harmonic_diagnostics.csv`.

Series	Statistic	Order	Held-out MSE	Chosen
`target`	`amplitude`	1	14.577	no
`target`	`amplitude`	2	13.248	yes
`target`	`phase`	1	3.581	yes
`target`	`phase`	2	3.583	no
`residual`	`amplitude`	1	2.880	no
`residual`	`amplitude`	2	2.805	yes
`residual`	`phase`	1	9.820	no
`residual`	`phase`	2	9.486	yes

The hour of the daily maximum is a circular quantity, and it was recentred on each series' circular mean before fitting, so that the two labels of midnight did not pull the fit apart. On the target the maximum sits near midnight throughout the year and its annual fit takes order one, with a held-out error that order two does not improve: the inclination's daily cycle keeps its timing, the wall tipping farthest towards the mountain in the afternoon and recovering overnight. On the residual the maximum sits near 17 h in winter and moves to about 23 h in July, an excursion of about six hours that order two fits better than order one by three per cent held out. What the drivers leave behind therefore changes its shape as well as its size through the year, which is the case for two weighted daily shapes rather than one scaled shape.

The diurnal band of the residual, binned by week of the year and slot of the day, is a surface whose singular values say how many weighted daily shapes it takes to reproduce it (Table 8). The first component carries 69 % of its variance and the second 24 %, 92.7 % together; a third adds 5 %. Two blended shapes therefore reproduce the surface to within a few per cent, which licenses the two conditions of D7 and does not call for a third.

Table 8: Singular-value decomposition of the residual's daily-by-annual surface: share of variance per component and cumulative share. Read from `GM_04_harmonic_diagnostics.csv`.

Component	Share	Cumulative
1	69.1%	69.1%
2	23.6%	92.7%
3	5.0%	97.8%
4	0.7%	98.4%
5	0.5%	99.0%
6	0.2%	99.1%

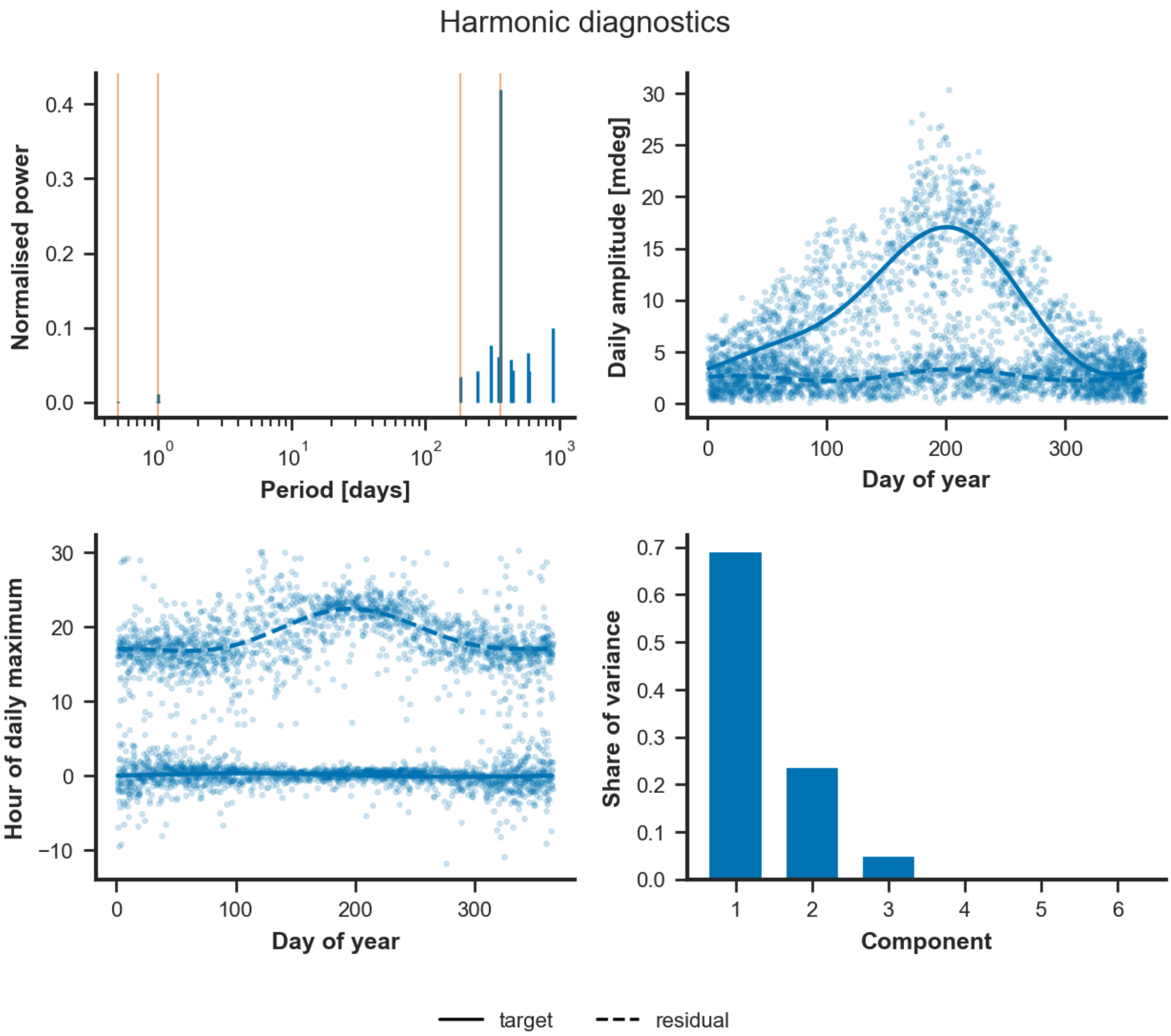

Figure 3: The harmonic diagnostic. Top left: the periods the scan certifies on the target (solid) and on the regression residual (dashed), on a logarithmic period axis, with the half-day, day, half-year and year marked. Top right: the daily amplitude by day of year with its annual fit. Bottom left: the hour of the daily maximum, recentred on its circular mean, with its annual fit. Bottom right: the share of variance of the leading components of the daily-by-annual surface.

Three parameters follow from the diagnostic and are declared in the notebook's parameter cell with a pointer to `GM_04`. The yearly term takes Fourier order one, because the residual certifies the annual cycle and no semi-annual one. The daily term takes order two, because both series certify the 12-hour companion of the daily cycle, the signature of a daily response that heats faster than it cools. The weight curve of the conditional daily term is the residual's order-two annual fit of the daily amplitude, normalised to the unit interval and peaking on day 206, and the two weighted shapes go forward to Phase 2's held-out test, since the first two components of the surface carry 92.7 % of its variance.

5.2 The fitted terms, and the conditional daily term's test

Figure 4 draws the two seasonal terms Model A fitted on the on-structure set. The yearly term, of Fourier order one, has a peak-to-peak of 44.4 mdeg and carries 12.2 % of the fitted variance (Table 10); on the station and ERA5 sets it is 51.9 and 49.3 mdeg and 16.8 and 14.6 %. On the on-structure set it reaches its maximum of +22.2 mdeg on 21 March, day 80, and its minimum of −22.2 mdeg on 20 September, day 263 (`GM_05d`): the term peaks at the equinoxes, in quadrature with the annual cycle of air temperature, which peaks near the solstices. That is the shape a delayed annual response leaves when its driver enters instantaneously, since the first-order correction for a lag is proportional to the driver's rate of change, largest when the season turns; the yearly term is the part of the wall's seasonal response that follows the air with a delay the instantaneous driver cannot supply. The delay itself is not estimated here. The daily term, of order two, is small, as D7 anticipated: a peak-to-peak of 3.9 mdeg and 0.1 % of the variance on the on-structure set, 8.2 mdeg and 0.4 % on the station set, 13.4 mdeg and 1.0 % on ERA5, larger where the driver is measured farther from the wall, since whatever the driver does not explain of the daily cycle is left for the term to hold. On the on-structure set it rises to +2.3 mdeg at 17:00 UTC and falls to −1.6 mdeg at 08:20 UTC (`GM_05d`), the evening and the morning, on either side of the afternoon maximum of the air temperature that carries the rest of the daily swing.

The conditional daily term did not earn its place (Table 9). Fitted as two shapes weighted by the measured annual curve, it raised the held-out error from 17.745 to 18.177 mdeg and carried 0.12 % of the variance against the plain term's 0.09 %, under the one per cent D7 set as the alternative criterion. Every fit downstream of this comparison therefore uses the plain daily term of order two, and the two shapes the design would have drawn at the solstices and equinoxes collapse to one curve, which is the one Figure 4 shows. The outcome is consistent with the diagnostic: once air temperature has been removed, the daily cycle's annual modulation is weak, 2.20 to 3.32 mdeg over the year (Table 7), and a term that varies with the season has almost nothing to modulate. D7's further fallback, four boolean seasons, is not reached: the smooth weights fitted; they were not worth their parameters.

Table 9: The plain daily term against the conditional one, on the on-structure set: held-out mean absolute error and the share of variance the daily term carries in each fit. Read from `GM_05b_conditional_test.csv`.

Daily term	Held-out MAE [mdeg]	Share of variance
`plain`	17.745	0.09%
`conditional`	18.177	0.12%

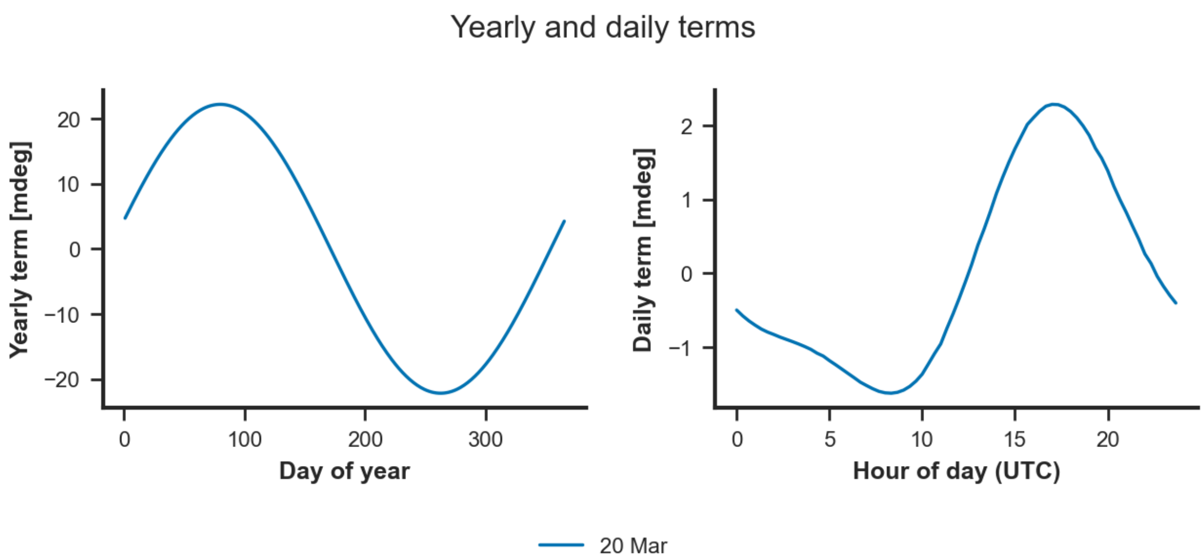

Figure 4: The seasonal terms of Model A on the on-structure set. Left: the yearly term over one synthetic year, by day of year. Right: the daily term over one synthetic day, by hour of day in UTC. With the conditional term rejected, the daily curve is the same on every day of the year.

6 Regressors and gains

Table 10 gives what each set's record is made of, and Figure 6 draws the decomposition of the on-structure set over the training window. The picture is the same on all three sets. The trend holds three fifths of the fitted variance (60.1, 60.7 and 64.1 %); air temperature holds the next largest share, 24.1 % on the on-structure set, 18.3 % on the station set and 15.6 % on ERA5, with a peak-to-peak contribution of 115.4, 100.9 and 105.0 mdeg; the yearly term holds 12 to 17 %; the residual 3.1, 3.7 and 4.2 %; and humidity, the daily term and radiation together hold 0.42 and 0.49 % on the two nearer sets and 1.51 % on ERA5. Net of its slow movement, the record is a thermometer: the contribution of air temperature alone swings by more than a hundred millidegrees.

Table 10: The share of the fitted variance each additive component carries and its peak-to-peak contribution over the training window, per regressor set, ordered by share. The on-structure set's radiation is the station's column, borrowed. Read from `GM_05_component_shares.csv`.

Set	Component	Share	Peak-to-peak [mdeg]
`str`	`trend`	60.1%	116.4
`str`	`future_regressor_tair`	24.1%	115.4
`str`	`season_yearly`	12.2%	44.4
`str`	`residual`	3.1%	60.6
`str`	`future_regressor_rh`	0.3%	8.8
`str`	`season_daily`	0.1%	3.9
`str`	`future_regressor_sr`	0.1%	5.8
`gs`	`trend`	60.7%	119.9
`gs`	`future_regressor_tair`	18.3%	100.9
`gs`	`season_yearly`	16.8%	51.9
`gs`	`residual`	3.7%	73.2
`gs`	`season_daily`	0.4%	8.2
`gs`	`future_regressor_rh`	0.1%	8.8
`gs`	`future_regressor_sr`	0.0%	0.6
`era5`	`trend`	64.1%	139.5
`era5`	`future_regressor_tair`	15.6%	105.0
`era5`	`season_yearly`	14.6%	49.3
`era5`	`residual`	4.2%	88.0
`era5`	`season_daily`	1.0%	13.4
`era5`	`future_regressor_sr`	0.3%	10.0
`era5`	`future_regressor_rh`	0.3%	11.5

6.1 The gains against Study 03

Table 11 and Figure 5 put each learned gain beside Study 03's. NeuralProphet's additive future regressor is linear in the driver, so the gain is the slope of the component against the driver and is read off exactly; the coefficient of determination of that reading is one by construction and is not tabulated. On the on-structure set the air-temperature gain is −2.639 mdeg per degree against Study 03's −2.79, within 5.4 % of it and inside the month-to-month range of −3.63 to −1.88 that Study 03 measured over 35 months. This is the agreement Checkpoint 2 required before the study could go on: an additive model with a trend, an annual term and three instantaneous drivers, fitted on six years of levels, recovers the coupling that Study 03 measured by a different method on the diurnal band. The station and ERA5 sets learn -2.402 and -2.295 against Study 03's -2.23 and -2.04, larger in magnitude by 7.7 and 12.5 %; a driver measured farther from the wall is smoother than the wall's own air, and a fit on levels compensates with a larger slope where Study 03's band-limited fit did not have to.

Relative humidity, for which Study 03 reports a positive correlation and no gain, learns $+0.108$, $+0.092$ and $+0.140$ mdeg per percentage point: positive on every set, small, and worth 8.8 to 11.5 mdeg peak-to-peak over the humidity range the record spans. Study 03 read the correlation as the inverse of the temperature cycle rather than as a mechanism, and nothing here contradicts that. Radiation is the one driver the model does not recover. Study 03 measured -0.035 mdeg per $\mathrm{W\,m^{-2}}$ for the station's radiation and -0.026 for ERA5's on the diurnal band; here the learned gains are $+0.006$ on the on-structure set, -0.001 on the station set and -0.011 on ERA5, with peak-to-peak contributions of 5.8, 0.6 and 10.0 mdeg, and on the on-structure set the sign is the wrong one. The discrepancy is less a contradiction of Study 03 than a statement about what a fit on levels can separate: radiation, delayed by an hour, moves in step with the air temperature that heats with it, and its marginal contribution once temperature and the daily term have taken theirs is close to zero. Study 03 asked what radiation does to the diurnal swing; this fit asks what it adds to the level beyond air temperature, and answers that it adds little. Section 7 returns to the question with a learned impulse response, and the ladder of Section 9 asks it again of the wall's own pyranometer.

Table 11: The gain each future regressor learned, per set, beside Study 03's diurnal-band measurement where one exists. Units: millidegrees per degree Celsius for air temperature, per percentage point for relative humidity and per $\mathrm{W\,m^{-2}}$ for radiation. Read from `GM_06_learned_gains.csv`.

Set	Driver	Learned gain	Study 03
`str`	`tair`	-2.639	-2.790
`str`	`rh`	0.108	—
`str`	`sr`	0.006	-0.035
`gs`	`tair`	-2.402	-2.230
`gs`	`rh`	0.092	—
`gs`	`sr`	-0.001	-0.035
`era5`	`tair`	-2.295	-2.040
`era5`	`rh`	0.140	—
`era5`	`sr`	-0.011	-0.026

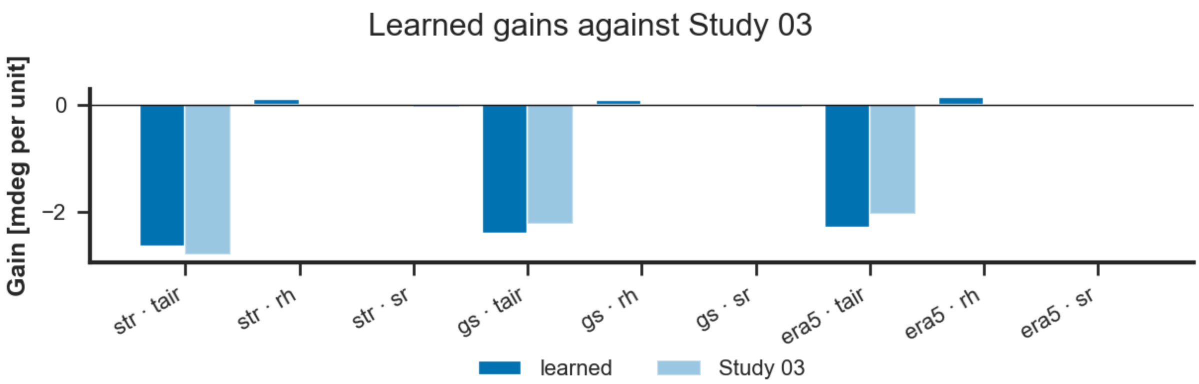

Figure 5: The learned gain per set and driver (solid) beside Study 03's measurement (light), in millidegrees per unit of the driver. Study 03 has no gain for relative humidity.

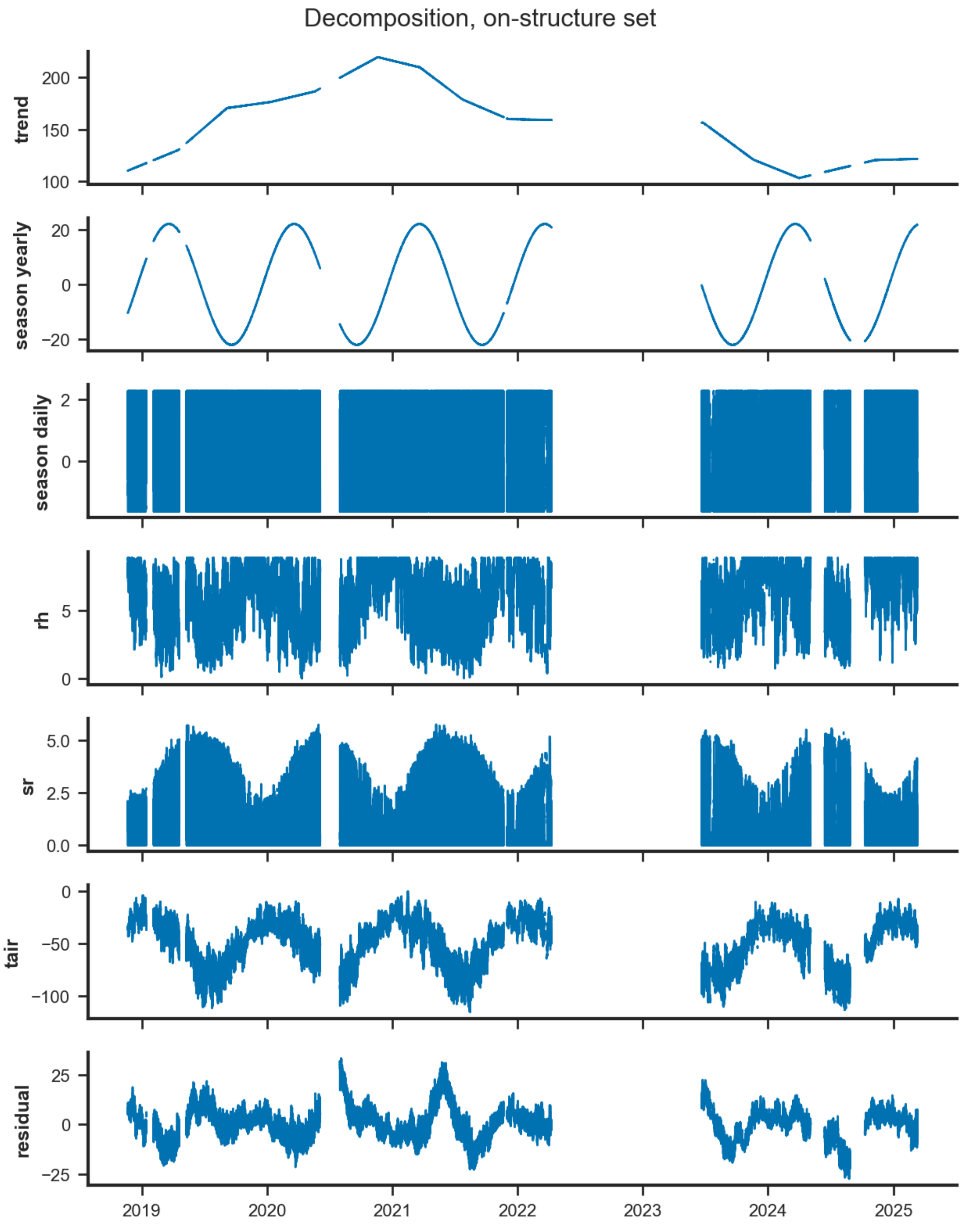

Figure 6: The additive decomposition of the on-structure set over the training window: the trend, the yearly and daily terms, the contribution of each driver and the residual, on one clock, with the lines broken across the gaps.

6.2 Stability across folds

Table 12 reports the gain, the yearly amplitude and the mean trend rate on each of NeuralProphet's five chronological folds, each refitted from scratch on the record up to its own held-out tenth. The air-temperature gain keeps its sign in every fold and stays within −2.45 to −2.70 on the on-structure set, −2.28 to −2.50 on the station set and −2.21 to −2.34 on ERA5; the yearly peak-to-peak stays within 39 to 45, 44 to 52 and 45 to 50 mdeg. Both are findings by the rule of Section 3. The mean trend rate is not a finding about the wall: it changes sign between folds on the on-structure and station sets and, on ERA5, where it keeps its sign, runs from 4.4 to 20.2 mdeg per year, because each fold's window ends at a different point of a record whose slow movement reverses; it is the mean of that record over the stretch a fold covers, not a rate of the structure. The held-out error of the folds, from 7.8 to 100.7 mdeg on the on-structure set, measures how far a trend extrapolated over a tenth of the set's rows, 13,657 slots of twenty minutes or about 190 days, wanders from the level the wall actually reached. That number is the case for the thirty-day refit of D5, which Section 8 scores.

Table 12: Component stability across NeuralProphet's five chronological folds, per set: the air-temperature gain, the yearly term's peak-to-peak, the mean trend rate and the held-out mean absolute error of each fold. Read from `GM_07_component_stability.csv`.

Set	Fold	Gain [mdeg/°C]	Yearly p–p [mdeg]	Trend rate [mdeg/yr]	Held-out MAE [mdeg]
`str`	1	-2.446	40.1	12.0	100.69
`str`	2	-2.523	39.4	-4.1	52.39
`str`	3	-2.500	42.9	-2.0	52.49
`str`	4	-2.639	44.7	-1.0	7.84
`str`	5	-2.701	43.8	3.1	21.06
`gs`	1	-2.282	43.6	11.6	106.38
`gs`	2	-2.294	45.0	-3.4	54.52
`gs`	3	-2.296	46.0	-0.6	44.52
`gs`	4	-2.402	52.0	-2.4	15.43
`gs`	5	-2.496	45.9	5.0	20.45
`era5`	1	-2.213	45.3	20.2	55.66
`era5`	2	-2.209	47.6	13.4	16.17
`era5`	3	-2.228	47.7	4.4	34.69
`era5`	4	-2.289	49.6	7.8	14.19
`era5`	5	-2.340	47.0	8.7	15.89

6.3 What the residual still holds

Table 14 gives the residual's scale and whiteness. Its standard deviation is 7.97 mdeg on the on-structure set, 8.65 on the station set and 9.47 on ERA5, with median absolute deviations of 4.76, 5.27 and 5.39: the nearer the driver, the less is left. It is not white. The Ljung–Box test rejects independence at lags of one slot, one day and three days on every set with a p-value that rounds to zero, and the bottom panel of Figure 6 shows why: the residual wanders over weeks, with a peak-to-peak of 60.6 mdeg on the on-structure set (Table 10). The component that would absorb this is the record's own memory, an autoregressive term, and D7 leaves it out on purpose, because with it the decomposition stops reading as physics. The memory is handed instead to the two places built for it: the rolling refit of D5, which re-anchors the expectation every thirty days so that the wandering never accumulates, and the prewhitening of D10, which removes the residual's lag-one dependence before a chart is drawn on it.

Table 13 names the periodic structure the residual still carries, scanned band by band as the diagnostic of Section 5.1 was. In the short band the daily line survives on every set, at normalised powers of 0.004 to 0.006, and it survives split in two: at 0.9972 and 1.0027 days on the on-structure and station sets, the first annual sidebands of the daily cycle, $1/(1 \pm 1/365.25)$ days, and at 0.9946 and 1.0027 days on ERA5, whose lower line is the second sideband, $1/(1 + 2/365.25)$ days. These are the seasonally modulated part of the daily cycle, which a daily term of fixed shape cannot remove. This is exactly the structure the conditional daily term was designed for, and Section 5.2

found that the term did not lower the held-out error; what remains is small, known, and left in the residual on purpose. The 12-hour line survives on the on-structure and ERA5 sets at 0.002 to 0.003. In the long band the largest peak on every set is semi-annual, 181 to 183 days at powers of 0.17 to 0.24, followed by entries at 325, 247, 220 and 150 days on the on-structure set. The semi-annual peak is a finding to be read with care. The yearly term's order was fixed at one by the diagnostic of D6, on a regression residual that certified no semi-annual cycle; the residual of the full model, after a trend with twelve changepoints whose segments are four to six months long, carries one. Whether it is a component of the wall's annual response that the order-one term cannot represent, or the imprint of the changepoint spacing on what the trend leaves behind, this study does not decide: by D6's rule the order is never raised beyond what the diagnostic scan supports, and the question is carried to Section 13.

Table 13: Periods certified by the Lomb–Scargle scan on the attribution fit's residual, per set, each band ranked against its own continuum, as in Table 6. Read from `GM_08b_residual_periods.csv`.

Set	Band	Rank	Period [days]	Power
`str`	`short`	1	0.9972	0.0042
`str`	`short`	2	1.0027	0.0027
`str`	`short`	3	0.4993	0.0022
`str`	`short`	4	0.9947	0.0019
`str`	`short`	5	1.0040	0.0010
`str`	`long`	1	181.1427	0.2449
`str`	`long`	2	324.8799	0.2358
`str`	`long`	3	247.1493	0.1994
`str`	`long`	4	219.9155	0.1097
`str`	`long`	5	150.2754	0.0899
`gs`	`short`	1	0.9972	0.0054
`gs`	`short`	2	1.0027	0.0047
`gs`	`short`	3	1.0016	0.0021
`gs`	`short`	4	0.9947	0.0020
`gs`	`short`	5	1.0003	0.0016
`gs`	`long`	1	181.6054	0.2426
`gs`	`long`	2	326.3473	0.1977
`gs`	`long`	3	248.7093	0.1419
`gs`	`long`	4	149.8416	0.1019
`gs`	`long`	5	275.5209	0.0926
`era5`	`short`	1	0.9946	0.0063
`era5`	`short`	2	1.0027	0.0055
`era5`	`short`	3	0.4993	0.0028
`era5`	`short`	4	0.9935	0.0022
`era5`	`short`	5	0.9989	0.0016
`era5`	`long`	1	182.7673	0.1654
`era5`	`long`	2	272.3530	0.1525
`era5`	`long`	3	215.3814	0.1019
`era5`	`long`	4	153.7001	0.0993
`era5`	`long`	5	241.4781	0.0793

Table 14: Residual diagnostics of the attribution fit, per set: the Ljung–Box p-value at lags of one slot, one day and three days, and the residual's standard deviation and median absolute deviation. Read from `GM_08_residual_diagnostics.csv`.

Set	Lag [slots]	Ljung–Box p	Std [mdeg]	MAD [mdeg]
`str`	1	0	7.97	4.76
`str`	72	0	7.97	4.76
`str`	216	0	7.97	4.76
`gs`	1	0	8.65	5.27
`gs`	72	0	8.65	5.27
`gs`	216	0	8.65	5.27
`era5`	1	0	9.47	5.39
`era5`	72	0	9.47	5.39
`era5`	216	0	9.47	5.39

7 Impulse response

Model A takes the coupling's timing from Study 03 and imposes it: air temperature enters instantaneously, radiation delayed by one hour, neither with a time constant. Model B is the same on-structure fit with that choice removed. Air temperature and radiation enter as lagged regressors over thirty-six slots of the native twenty-minute grid, twelve hours of history per driver, while relative humidity stays contemporaneous and the trend, the yearly and daily terms, the changepoints and the learning rate are those of the attribution fit. The weight the fit learns at each lag is the driver's impulse response in millidegrees per unit of the driver, and three numbers are read from it: the gain, which is the sum of the weights; the delay, which is the weight-weighted mean lag, the discrete counterpart of the transport delay Study 03's operator applies as a whole-sample shift; and the time constant τ of the one-pole step response fitted to the response's cumulative sum, with the coefficient of determination of that fit. Table 15 sets the three beside Study 03's cell for each driver, and Figure 7 draws the weights with Study 03's operator overlaid as a gain-matched response.

Table 15: The impulse response Model B learned on the on-structure set, reduced to a gain, a delay and a time constant per driver, beside the delay and time constant Study 03 measured and Model A imposes. The gain is the summed weight in millidegrees per degree Celsius for air temperature and per $\mathrm{W\,m^{-2}}$ for radiation; the delay is the weight-weighted mean lag; τ and R^2 are the one-pole step response fitted to the cumulative weight and how much of the cumulative weight's variance it explains, negative when it explains less than the mean does. Read from `GM_10_impulse_response.csv`.

Driver	Gain	Delay [h]	τ [h]	R^2 one-pole	Study 03 delay [h]	Study 03 τ [h]
`sr`	0.023	6.50	8.00	0.929	1.0	0.0
`tair`	-2.668	1.44	0.50	-0.466	0.0	0.0

For air temperature the learned response agrees with the imposed one, and the table's delay column has to be read with the weights in view to see it. The gain is -2.668 mdeg per degree, within 4.4 % of Study 03's -2.79 and within 1.1 % of the -2.639 Model A learned with the driver entering instantaneously (Table 11). Of that gain, -1.856 mdeg per degree, 69.6 %, sits in the first lag, the twenty minutes before the observation, and -2.283, 85.6 %, in the first hour; the remaining thirty-three weights sum to -0.384, alternate in sign and none exceeds 0.276 in magnitude. The 1.44 h in the delay column is the first moment of that whole sequence, and 1.08 h of it comes from the small alternating weights between one and twelve hours, against 0.36 h from the first hour that holds six sevenths of the gain. The one-pole fit says the same thing in its own way: the cumulative weight reaches −1.856 at the first slot and then stays between −1.84 and −2.77 over the remaining lags, a step and not an exponential rise; the best one-pole time constant is 0.5 h, and even that shape explains the cumulative weight worse than its own mean does, hence the negative R^2 of −0.466. At twenty minutes the wall's air-temperature response has no resolvable delay and no time constant, which is Study 03's cell of zero and zero, and the operator Model A imposes is the one Model B finds.

For radiation the learned response has no shape at all. The gain is +0.023 mdeg per $\mathrm{W\,m^{-2}}$, positive as Model A's +0.006 on this set was and opposite in sign to the −0.035 Study 03 measured on the diurnal band; the correction of the weights into physical units cannot have produced the sign, since it multiplies every weight by a ratio of two positive scales. No lag stands out. The largest weight, 0.0064, is at the second lag, forty minutes before the observation; the next two, 0.0049 and 0.0038, are at the last lag and at six hours forty; twenty-four of the thirty-six weights are positive and twelve negative; the weight at one hour, the lag at which Study 03 places the response, is −0.0026. The first hour holds 9.4 % of the gain and the weights beyond six hours as much as the weights within them, 0.0111 against 0.0118, so the first moment falls at 6.5 h, the middle of the window, where the first moment of any sequence without a preferred lag falls. The one-pole fit reports $\tau = 8.0$ h with $R^2 = 0.929$, and this is the one number in the table that has to be disbelieved: the cumulative sum of thirty-six small weights of the same average sign climbs almost linearly from 0.002 at one hour to 0.023 at twelve, and a line across a twelve-hour window is what a one-pole response with a time constant of the window's order draws. The fit certifies the drift of the weights, not a response of the wall. Model B therefore neither confirms nor refutes the one-hour radiation delay Study 03 measured. It repeats the finding of Section 6: once air temperature has taken its share of the level, the marginal weight of the radiation that heats with it is indistinguishable from noise at every lag the grid resolves, and the sign of its sum is a finding stated, not a coupling recovered.

Checkpoint 4 is therefore passed for air temperature by agreement and left open for radiation by the absence of any response to compare. Model A is not re-tuned on either count. The one-hour radiation delay stays imposed, as D9 provides, since the evidence for it is Study 03's diurnal-band scan and the evidence against it is none; and the instantaneous air-temperature coupling is now validated at the resolution Study 03's hourly scan could not reach.

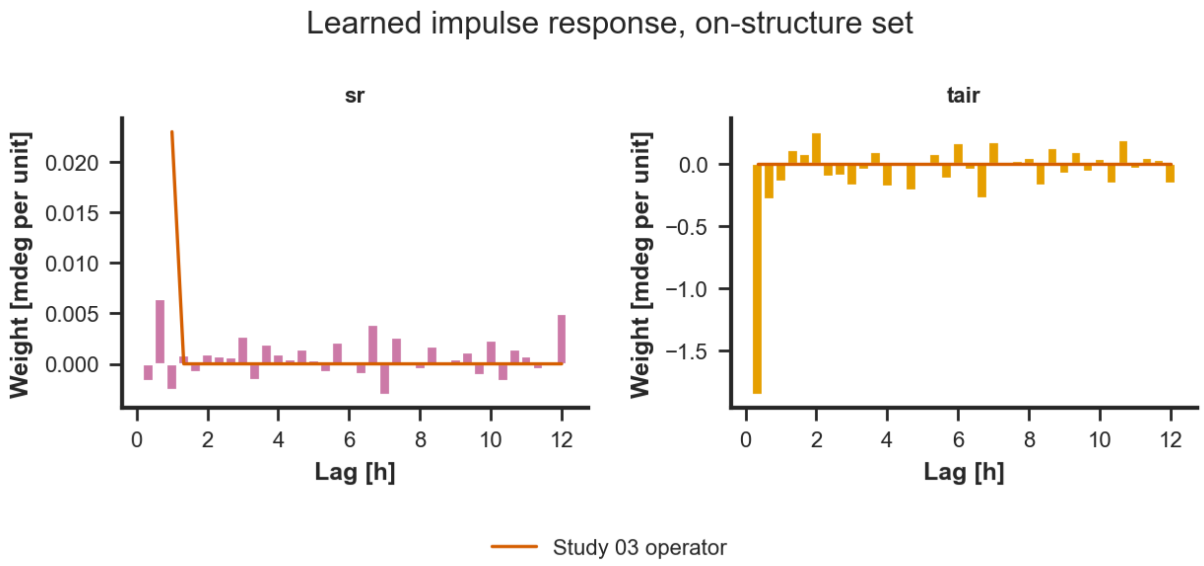

Figure 7: The impulse response Model B learned on the on-structure set, one panel per driver: the weight at each of the thirty-six lags of the twenty-minute grid, in millidegrees per unit of the driver, with Study 03's operator overlaid as a response of the same gain. The air-temperature response is one slot deep; the radiation response has no lag that stands above its neighbours.

8 Uncertainty and validation

Three things are validated here: that the attribution fit converged and where its held-out error settles; that its components hold across the library's chronological folds; and that the rolling expectation of D5, with the rolling conformal interval of D8, is accurate and covers at the rate it claims, as a function of how stale the last refit is.

8.1 The fit and the folds

Figure 8 draws the library's own training curve for the on-structure attribution fit, the mean absolute error on the training rows and on the held-out last fifth at every epoch; the values behind it are in `GM_05e_fit_metrics.csv`. The training error falls through the thirty epochs and settles at 6.1 mdeg; the held-out error reaches its minimum of 11.5 mdeg after six epochs and then rises to the 17.7 mdeg at the thirtieth that Table 4 reports for the same fit. That divergence is not a defect of the optimiser: the held-out rows are the last fifth of a level record, from March 2025 onward, and a model fitted once on the years before them extrapolates its last trend segment across them. The same thing, measured across five folds, is what Table 12 showed in Section 6: the gain and the yearly amplitude are stable, the held-out error of a fold is not, and it ranges from 7.8 to 100.7 mdeg on the on-structure set according to where the fold's window ends. A single fit cannot be the expectation; the expectation has to be refitted as the record grows, which is D5.

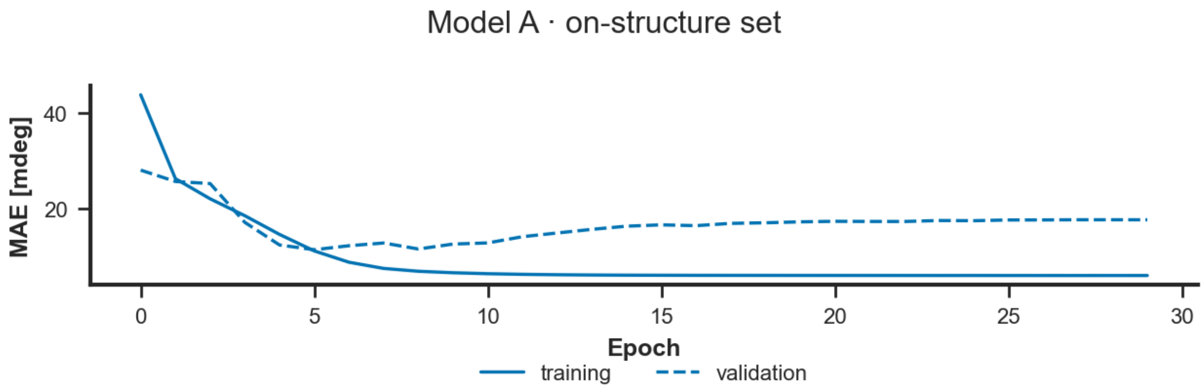

Figure 8: The training curve of the on-structure attribution fit: mean absolute error on the training rows (solid) and on the held-out last fifth (dashed) by epoch, in millidegrees.

8.2 The rolling expectation, by days since refit

The expectation was refitted every thirty days from 20 November 2020, the first origin with two annual cycles of history behind it, to the end of the record, each refit on the trailing three years with its changepoints placed on the covered time of that window, and each refit's interval calibrated by split conformal prediction on the previous six months of its own out-of-sample residuals (D5, D8). Every row of the record from that origin onward is therefore predicted by a model that has not seen it, from an origin at most thirty days old. Table 16 scores the result per regressor set and per week of staleness: 91,187 rows on the on-structure set, 88,414 on the station set and 100,583 on ERA5.

Accuracy decays with staleness, and it decays the same way on every set. On the on-structure set the mean absolute error is 5.31 mdeg in the first week after a refit, 7.54 in the second, 9.05 in the third and 10.89 in the fourth, a growth of about 1.9 mdeg per week; the station set runs from 5.90 to 12.09 and ERA5 from 6.77 to 13.08. The bias stays within ±1.2 mdeg in every bin. Pooled over the month, the mean absolute error is 8.40 mdeg on the on-structure set, 9.38 on the station set and 10.19 on ERA5: the nearer the driver, the better the expectation, by about one millidegree per step outward. The growth within the month is larger than the trend line's own extrapolation error, which D5 estimated at a fraction of a millidegree over thirty days; what accumulates between refits is the residual's slow wandering, the memory Section 6 found in the residual and D7 kept out of the model, which each refit re-anchors and the next month lets drift again.

Coverage decays with staleness too, and this is the interval's finding. The nominal 90 % interval covers 96.2 % of the on-structure rows in the first week, 90.2 % in the second, 87.3 % in the third and 81.9 % in the fourth; the station and ERA5 sets run from 97.1 to 81.0 % and from 95.5 to 78.1 %. Its width does not change with staleness: about 35 mdeg on the on-structure set, 39 on the station set and 46 on ERA5 in every bin, because the calibration pools the previous six months of residuals across every staleness and so fixes one width for the whole month. The interval is therefore right on average and wrong at both ends, too wide in the week after a refit and too narrow in the week before the next. Pooled over the month it covers 88.4 % of the on-structure rows, 88.0 % of the station set's and 86.5 % of ERA5's, against the nominal 90 % and against the 67.7 % that Study 04 reached with an interval calibrated once, eleven months before the record it was asked to cover. Two remedies follow from the table and are not applied here: a calibration conditioned on staleness, which would let the width grow through the month, or a shorter refit interval, which would cut the month off before the fourth week; either is a design choice for the deployment rather than for this study, and the interval score of the last column, which charges both width and misses, is the number on which to compare them.

Table 16: The rolling expectation scored per regressor set and per week since the last refit: rows, mean absolute error, root mean square error and bias in millidegrees, the coverage and mean width of the nominal 90 % conformal interval, and the interval score. Read from `GM_09_nowcast_metrics.csv`.

Set	Staleness	Rows	MAE	RMSE	Bias	Coverage	Width	Interval score
`era5`	`0-7 d`	23,185	6.77	8.90	0.77	95.5%	46.6	56.4
`era5`	`8-14 d`	23,207	9.28	13.05	0.88	89.4%	46.2	75.5
`era5`	`15-21 d`	23,886	10.75	14.99	0.44	85.4%	45.4	85.1
`era5`	`22-30 d`	30,305	13.08	17.08	-0.44	78.1%	45.0	100.0
`gs`	`0-7 d`	20,349	5.90	7.92	0.36	97.1%	38.6	58.7
`gs`	`8-14 d`	20,362	8.42	11.95	1.09	90.0%	38.8	74.4
`gs`	`15-21 d`	20,746	10.22	14.54	1.13	86.1%	39.1	87.8
`gs`	`22-30 d`	26,957	12.09	15.98	-0.11	81.0%	38.7	98.5
`str`	`0-7 d`	21,219	5.31	7.22	0.31	96.2%	35.9	50.9
`str`	`8-14 d`	20,646	7.54	10.61	0.86	90.2%	34.9	66.1
`str`	`15-21 d`	21,603	9.05	12.73	1.17	87.3%	36.9	75.0
`str`	`22-30 d`	27,719	10.89	14.31	0.02	81.9%	34.5	81.1

Figure 9 shows one month of the on-structure expectation against the record, with its interval: December 2025, the first full month after the autumn 2025 outage in which every driver of the set is present, since the station's radiation is missing for most of November. The daily cycle of the observed inclination is followed in phase and in size, the expected level sits a few millidegrees above the observed one for most of the month and the interval covers it throughout: a departure of that shape and size, sustained across a refit month, is exactly what the daily and slow charts of Section 10 are built to judge, and the interval, which asks only whether a reading is plausible given the last refit, is not the instrument for it.

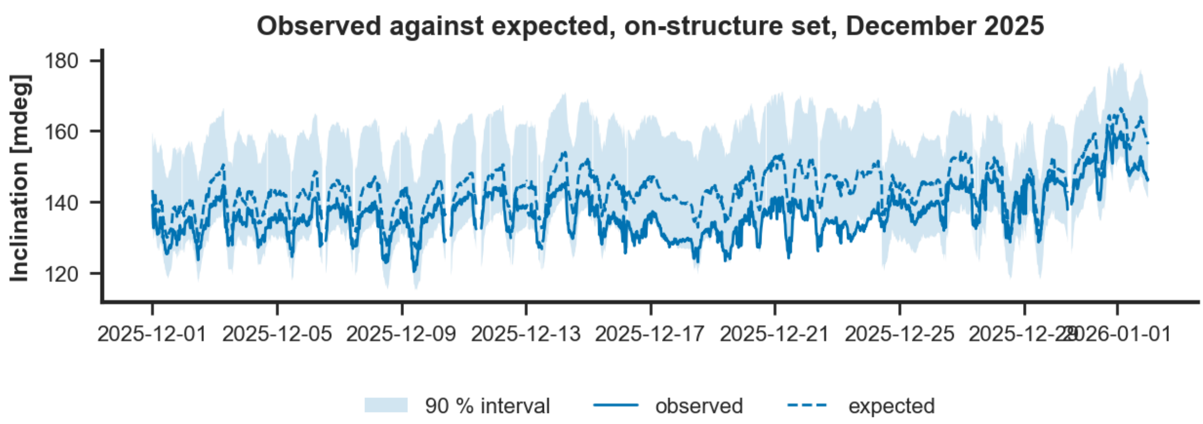

Figure 9: Observed inclination against the rolling expectation with its 90 % conformal interval, on-structure set, December 2025, on the twenty-minute grid.

9 What the wall temperature and the pyranometer buy

The probe and the pyranometer were installed on 21 February 2025 to find out whether they carry information the other drivers do not, and D4 tests them as a ladder on the current era alone. The window is the one on which every channel exists at once: the probe's two valid blocks and the pyranometer's certified days, 13,914 slots of twenty minutes, about 193 days. Every rung is fitted and scored on those rows, so that a rung differs from the one below in its channels alone; the first rung, which reads the station's radiation instead of the wall's, holds 13,290 of them, the difference being the station's own dropouts inside the window. Each rung fits Model A, with the plain daily term and the parameters of Section 4, on the first four fifths of its rows and scores the last fifth. Two cautions govern the reading. The comparison between rungs is the paired block-bootstrap skill on their shared held-out timestamps, not the mean absolute error column, which is each rung's own held-out fifth and, for the first rung, a slightly different stretch. And the ladder is the one place in this study where a model is fitted on less than two annual cycles, by D4's design: its gains compare channels against each other on one window and are not estimates of the wall's parameters; those are Section 6's.

Table 17 and Figure 10 give the result, rung by rung. Replacing the station's radiation with the wall's own pyranometer, on the same rows, raises the held-out error by 2.3 % (bootstrap bounds 1.6 to 3.2 %). The pyranometer's learned gain is -0.002 mdeg per $\mathrm{W\,m^{-2}}$, against the -0.026 Study 03 measured for it on the diurnal band and the -0.008 the station's column learns on the rung below; Section 2 had already found that its clock fails Study 02's agreement test. Adding the wall temperature at zero delay with a four-hour time constant, the deployable form, changes nothing over the rung below (skill 0.000, bounds -0.3 to $+0.4$ %) and leaves the rung 2.2 % below the first (bounds 1.2 to 3.3 %). Its learned gain is -0.30 mdeg per degree, a sixth of the -1.75 Study 03 measured on the diurnal band: in a fit on levels that already carries the air temperature and a trend, the probe's slow component is shared with both, and what remains for it is small. The residual's lag-one autocorrelation falls down the ladder, from 0.827 on the first rung to 0.801, 0.798 and 0.789, so each channel absorbs a little of the residual's memory without lowering its size. The diagnostic fourth rung, the probe at its measured lead of two hours, which reads the future and can never serve an expectation, improves on the third by 1.3 % (bounds 0.7 to 1.9 %) with a gain of -0.68 mdeg per degree, and still does not reach the first rung (skill against it -1.1 %, bounds -2.9 to $+0.5$ %). The coverage column is the raw quantile interval of a model fitted once. The held-out fifth of the matched rows, 2,783 slots or about 39 days on rungs 2 to 4 and 2,658 slots on the first, is the tail of the probe's second valid block, 18 June to 10 August 2026, so the fit that must hold there was made on the first block, of 2025, and the opening two weeks of the second, across the eleven months in which the probe recorded nothing. At 10 to 27 % against a nominal 90 % the raw interval says what Study 04 found and D8 corrects, and it is not the study's interval, which Section 8 calibrates.

The verdict per channel, in D4's terms. The pyranometer: a deployment gains nothing from it for the expectation, on this window; the station's radiation, an hour away, serves the model at least as well, and the wall's own instrument earns its place only if something other than the expectation wants it. The wall-temperature probe: in its deployable form it buys nothing measurable over the drivers the model already has, and its information about the wall arrives two hours before the deformation, where a monitor cannot use it; a deployment should not count on it for the expectation, and Study 03's finding that it is an internal state variable rather than a driver stands.

Table 17: The current-era ladder, on the matched window from 21 February 2025: rows fitted and scored, held-out mean absolute error of the last fifth, the paired block-bootstrap skill over the rung below and over the first rung with their 5 and 95 % bounds, the raw quantile interval's coverage of the held-out fifth, the learned gain of the rung's last channel (radiation in mdeg per $\mathrm{W\,m^{-2}}$, wall temperature in mdeg per degree) and the residual's lag-one autocorrelation. Read from `GM_16_current_era_ladder.csv`.

			Skill over rung below			Skill over rung 1					
Rung	Rows	MAE	Skill	5 %	95 %	Skill	5 %	95 %	Cov.	Gain	r_1
`1 on-structure set`	13,290	17.04	—	—	—	—	—	—	26.7%	-0.008	0.827
`2 on-structure radiation`	13,914	21.34	-0.023	-0.032	-0.016	-0.023	-0.032	-0.016	15.8%	-0.002	0.801
`3 + twall, tau 4 h`	13,914	21.33	0.000	-0.003	0.004	-0.022	-0.033	-0.012	21.2%	-0.296	0.798
`4 twall at its lead (diagnostic)`	13,914	21.05	0.013	0.007	0.019	-0.011	-0.029	0.005	9.8%	-0.675	0.789

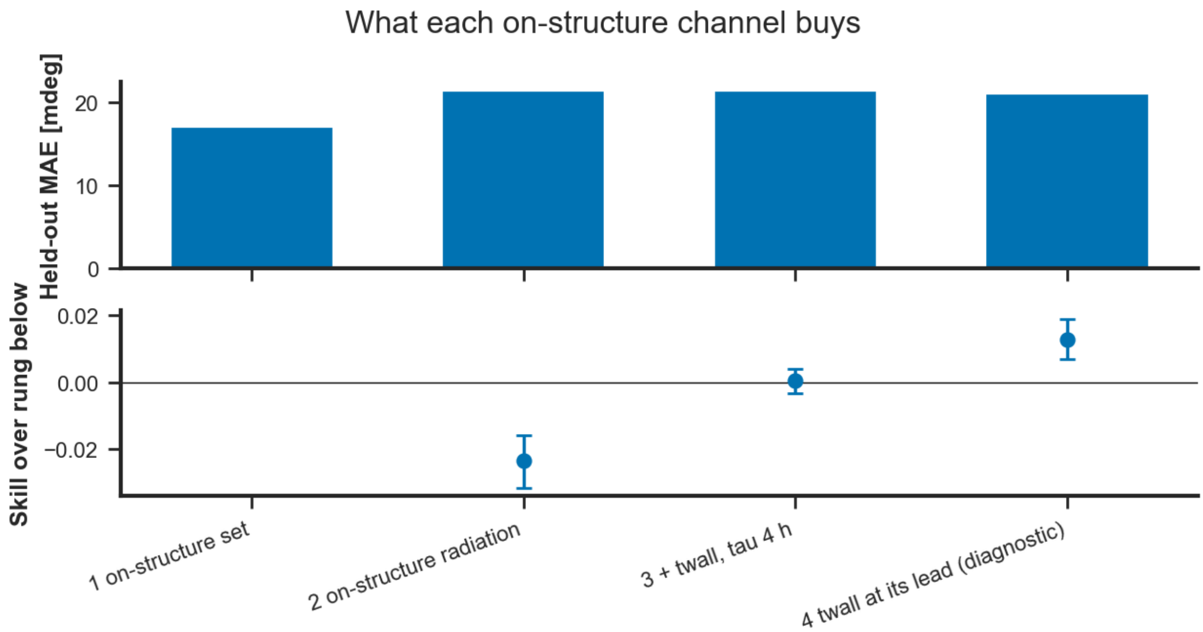

Figure 10: The current-era ladder. Top: held-out mean absolute error per rung. Bottom: the paired block-bootstrap skill of each rung over the one below, with its 5 and 95 % bounds; zero means the added or replaced channel changed nothing.

10 The monitor

This section is written after Phase 5. Nothing here is a result yet.

11 Outages as hypotheses

This section is written after Phase 6. Nothing here is a result yet.

12 Verdict

This section is written after Phase 7. Nothing here is a result yet.

13 Limitations

This section is written after Phase 7. Nothing here is a result yet.

14 Run metadata

This section is written after Phase 7. Nothing here is a result yet.